

Data Cleaning Essentials: Handling Missing Values and Outliers

A comprehensive guide to transforming raw, messy datasets into reliable foundations for machine learning and statistical analysis.



Introduction to Data Cleaning

The Golden Rule

"Garbage In, Garbage Out"
(GIGO)

Data quality determines the quality of your insights and predictions. No algorithm can compensate for fundamentally flawed data.

Core Challenges

- Missing values creating gaps in your dataset
- Outliers skewing statistical results and model performance

Objective: Transform raw, messy data into a reliable foundation for analysis.



Understanding Missing Values: The "Why" Matters

Identifying the pattern of missingness is crucial—ignoring the underlying cause can introduce significant bias into your models.

MCAR

Missing Completely at Random

No discernible pattern. The missingness is purely accidental and unrelated to any variables in the dataset.

MAR

Missing at Random

Missingness is related to other observed variables in your dataset, but not to the missing value itself.

MNAR

Missing Not at Random

The value is missing because of the value itself. Example: high-income earners deliberately hiding their salary.

Strategies for Handling Missing Values

01

Deletion

Listwise deletion: Remove the entire row from analysis. Best used when less than 5% of your data is missing.

02

Simple Imputation

Replace missing values with **mean** (average) or **median** (middle value for skewed data). For categorical data, use **mode** (most frequent category).

03

Advanced Imputation

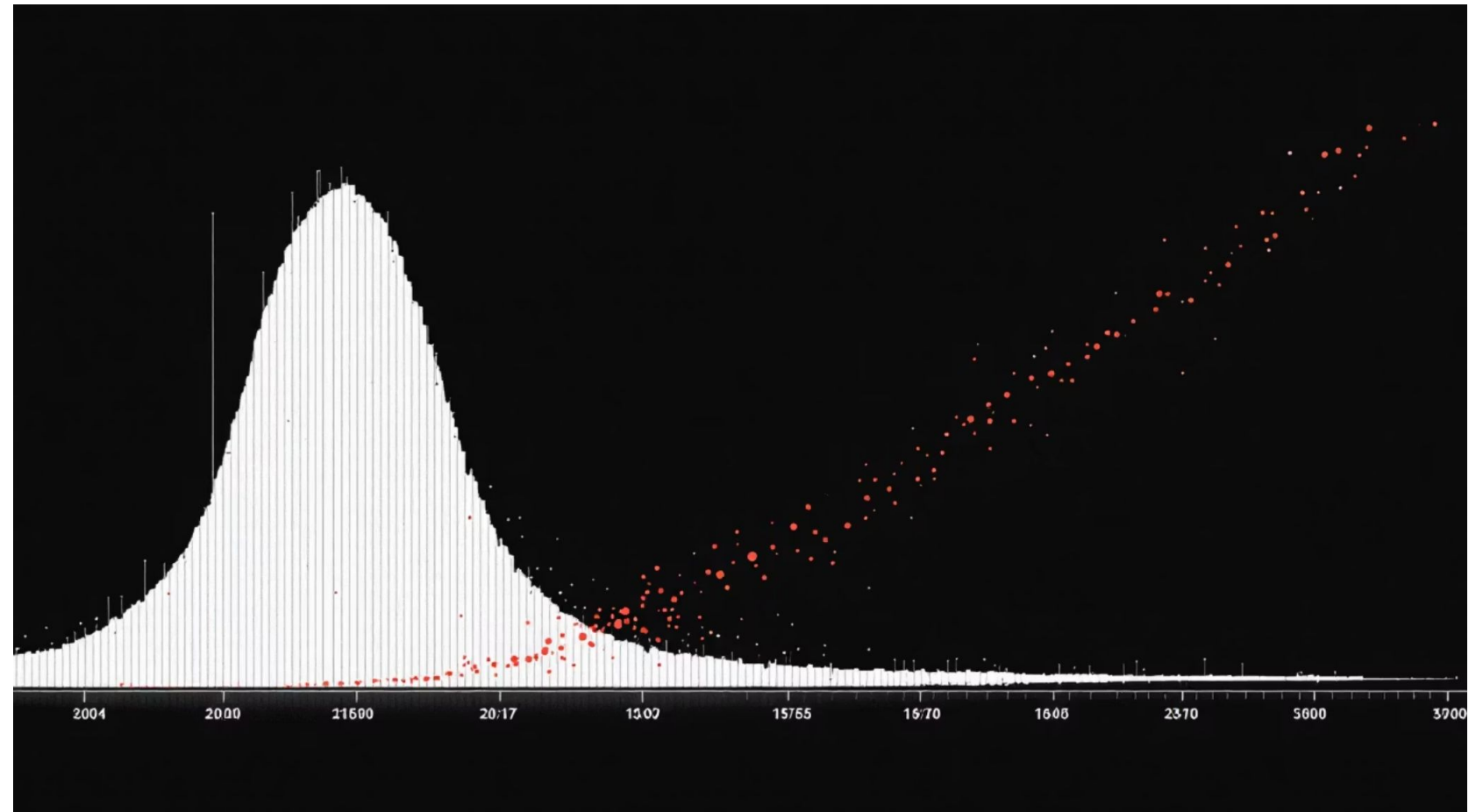
KNN (K-Nearest Neighbors): Fill gaps based on similar records. **MICE (Multivariate Imputation by Chained Equations):** Sophisticated iterative approach using multiple variables.



Detecting Outliers in Your Data

What Are Outliers?

Observations that deviate significantly from the rest of the distribution. They can dramatically impact statistical results and model performance.



Z-Score Method

Measures distance from the mean in standard deviations. A common threshold: $|Z| > 3$

IQR (Interquartile Range)

Measures spread between the 25th and 75th percentiles

Lower bound: $Q_1 - 1.5 \cdot IQR$

Upper bound: $Q_3 + 1.5 \cdot IQR$

Managing Outliers: Treatment Options

❏ Don't just delete! First, determine if the outlier is a measurement error or a rare but valid event.

1

Trimming

Remove extreme values. Use with caution as you lose potentially valuable information.

2

Winsorization

Set outliers to a specified percentile (e.g., the 95th percentile). Also called "capping."

3

Transformation

Apply log or square root transformations to compress the distance of outliers from the mean.

4

Imputation

Treat the outlier as a missing value and replace it with the median or another imputation method.

